

Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited training dataset	May reduce prediction accuracy for unseen data	High
2	Single ML model used	Performance may not be optimal for all cases	Medium
3	Basic security implementation	Vulnerable to unauthorized access if deployed publicly	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on cloud with load balancing and auto-scaling
2	Data Storage	Local database/storage	Migrate to cloud databases such as MySQL/PostgreSQL on AWS or Azure
3	Performance	Single-server execution	Optimize model, introduce caching, and use GPU/parallel processing if required
4	Security	Basic authentication	Implement JWT authentication, HTTPS, encryption, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 1	Improve model accuracy using larger and updated datasets	1 Month	Better prediction accuracy
Phase 2	Deploy application on cloud	2 Months	Improved scalability and accessibility
Phase 3	Add explainable AI (prediction reason) and dashboard	3 Months	Better transparency and user experience
Phase 4	Integrate with banking APIs for real-time verification	6 Months	Automated and faster credit card approval process